GM UNIVERSITY

COURSE DOCUMENT

Bachelor of Computer Application

GM UNIVERSITY
Innovating Minds

Faculty of Computing & Information Technology
School of Computer Application

Contents

SL NO	Course Code	Particular	Page No
		SEMESTER - I	
1	UL24FHK11	Kannada Shravana-1	3
2	UL24FHH11	Katha Saritha (Hindi)	4
3	UL24FHE11	Insight English-1	5
4	UC24CA1104	Fundamentals of Computer	6
5	UC24CA1105	Programming in C	7
6	UC24CA1106	C Programming Lab	8
7	UC24CA1107	Introduction to Linux	9
8	UC24CA1108	Linux Lab	10
9	UM24BC1142	OEC-1	11
		SEMESTER – II	
10	UL24FHK21	Kannada Shravana-2	12
11	UL24FHH21	Katha Sourabha (Hindi)	13
12	UL24FHE21	Insight English-2	14
13	UC24CA1204	Data Structure using C	15
14	UC24CA1205	Data Structure Lab	16
15	UC24CA1206	Object Oriented Concepts using Java	17
16	UC24CA1207	Java Lab	18
17	UM24BC1241	OEC-2	19
		SEMESTER – III	
18	UC24CA2301	Data Base System Concepts	20
19	UC24CA2302	Data Base System Lab	21
20	UC24CA2303	Python Programming	22
21	UC24CA2304	Python Lab	23
22	UC24CA2305	Operating System	24
23	UC24CA2306	Discrete Mathematics	25
24	UM24BC2441	OEC-3	26
		SEMESTER – IV	
25	UC24CA2401	Introduction to AI	27
26	UC24CA2402	Artificial Intelligence Lab	28
27	UC24CA2403	Design and Analysis of Algorithm Using C	29
28	UC24CA2404	Design and Analysis of Algorithm Lab	30
29	UC24CA2405	Computer Network	31
30	UC24CA2441 UC24CA2442 UC24CA2443	**Specialisation-1**	33-49

1.1 Kannada Shravana-1 (UL24FHK11)

Course Content

ಘಟಕ: 1 ನಾಡು–ನುಡಿಯ ಚಿಂತನೆ

1. ಕನ್ನಡ ಪದಗೊಳ್ – ಜಿ. ಪಿ ರಾಜರತ್ನಂ
2. ಚೋಮನ ದುಡಿ – ಡಾ. ಕೆ ಶಿವರಾಮ ಕಾರಂತ
3. ಮಂಕುತಿಮ್ಮನ ಕಗ್ಗ – ಡಿ ವಿ ಗುಂಡಪ್ಪ
4. ಇರುವುದೊಂದೇ ಭೂಮಿ – ನಾಗೇಶ ಹೆಗಡೆ

ಘಟಕ: 2 ಸಂಸ್ಕೃತಿ

5. ಹದಿಹರೆಯದ ವಿದ್ಯಾರ್ಥಿಗಳ ಸ್ಥಿತಿಗತಿಗಳು – ಡಾ. ಕೆ ಆರ್ ಶ್ರೀಧರ
6. ವಚನ ಸಾಹಿತ್ಯ ಆಯ್ದ ಭಾಗ – ಅಕ್ಕಮಹದೇವಿ, ಬಸವಣ್ಣ, ಜೇಡರ ದಾಸಿಮಯ್ಯ
7. ಬರಿದಾಗುತ್ತಿರುವ ನನ್ನ ಪ್ರೀತಿಯ ಕಡಲು – ಹೆಚ್ ನಾಗವೇಣಿ
8. ಕನ್ನಡಕ್ಕೊಂದು ಕಾರ್ಯಸೂಚಿ – ಬರಗೂರು ರಾಮಚಂದ್ರಪ್ಪ

ಘಟಕ: 3 ವ್ಯಾಕರಣಾಂಶಗಳು

ಗಾದೆಗಳು
ಪತ್ರಲೇಖನ
ಪ್ರಬಂಧ

1.2 गद्य:(UL24FHH13)

1) बदला –श्रीराम शमा

2) ममत्रता- आचार्ा रामचन्द्र शुक्ल

3) मबिंदा- महादेवी वमा

4) प्रदूषण-डाँ हरचरणलाल शमा

पद्य:

1) मप्रर्तम – सूर्ाकाि्ंत मत्रपाठी ' मिराला '

2) पुष्प की अमिलाषा – माखिलाल चतुवेदी

3) हमें ि बाँध ि्प्राचीर ि्में – डाँ. मशवमिंगल मसिंह 'सुमि'

4) पद – रैदास (द हा)

व्याकरण: विकारी शब्द

1) सिंजा : पररिाषा और प्रकार
2) सवाि्ाम: पररिाषा और प्रकार
3) मवशेषण: पररिाषा और प्रकार
4) मिर्ा : पररिाषा और प्रकार

* निबंध- लेखि

* अिुवाद लेखि

1.3 (UL24FHE12)

1. The Last leaf - O Henry
2. The Wolf -Farooq Sarwar
3. Relations between Men and Women -Rajaram Mohan Roy
4. Ignorance Isn't Bliss -Tabish Khair

POETRY

1. The Tables Turned -William Wordsworth
2. Where the Mind is Without Fear -Rabindranath Tagore
3. The Passionate Shepherd to His Love - Christopher Marlowe
4. Stopping by Woods on a Snowy Evening -Robert Frost

LANGUAGE PRACTICE

1. Articles
2. Preposition
3. Verbs
4. Tenses
5. Letter Writing
6. Comprehension Reading Prose

1.4 Fundamentals of Computer(UC24CA1104)

Anatomy of C omputer:

Definition of Computer, Characteristics of Computers, Generations of computer, Block Diagram of Digital Computer, Classifications of computers, CPU, Microprocessor, Storage units, Peripheral Devices.

Software Components and Problem Solving:

Define software, Classification of Software, levels of Languages - Machine Level, Assembly Level & High-Level Languages, Assembler, Translator-Interpreter and Compiler; Algorithm, Flowchart and symbols.

Number Systems:

Different types of Number systems, conversion from one number system to another; Computer Codes – BCD, Gray Code, ASCII, Boolean algebra – Boolean Operators with Truth Tables.

Operating System Fundamentals:

Operating Systems: Introduction, Functions of an operating System, Classification ofOperating Systems, Utilities, The Unix Operating System, Basic Unix commands, Difference between Windows and Unix operating systems.

Database Management System:

Define DB, DBMS, Why Database?, File system vs DBMS, Database applications,Database users, Roles of database admin, SQL – DDL and DML.

Internet:

Introduction, application of Internet, URL, Browser, types of Browsers, searchengines, Domain Names Services and types of domains, ISP.

1.5 Programming in C

Overview of C; History and Features of C; Structure of a C Program with Examples; Creating and Executing a C Program. C Programming Basic Concepts: C Character Set; C tokens - keywords, identifiers, constants, Symbolic constants, Data types; variables; Declaration & initialization of variables.

Input and Output functions, Operators and Expressions:

Formatted I/O functions– printf() and scanf() Unformatted I/O functions – getchar(), putchar(), gets() and puts() functions. Arithmetic operators; Relational operators; Logical operators; Assignment operators; Increment & Decrement operators; Bitwise operators; Conditional operator; Special operators; Operator Precedence and Associatively.

Control Structures, Arrays & Strings:

Decision making Statements- Simple if, if_else, nested if_else, else_if ladder, Switch Case, goto, break & continue statements; Looping Statements- while, do-while, for loops. Arrays: One & Multi-Dimensional arrays– Declaration and Initialization. Strings: Declaring & Initializing string variables; String handling functions- strlen(), strcmp(), strcpy(), strcat(), strrev(), toupper() and tolower().

Functions, Pointers, Structures and Unions:

Introduction to Functions, Function Declaration & Definition, Function call Mechanisms, Return values, Scope and lifetime of variables, Recursivefunctions, Categories of user defined functions. Pointers - Declaring and initializing pointers, accessing address and value of variables using pointers; Advantages and disadvantages of using pointers; Structures-Structure Definition, Declaring structure variables, Advantages of Structure, Unions– Union definition; Declaring union variables, Difference between Structures and Unions.

1.6 Programming Lab

1. Program to read radius of a circle and to find area and circumference
2. Program to find the factorial of a given number
3. Program to read three numbers and find the biggest of three
4. Program to generate Fibonacci Series
5. Program to generate n prime numbers
6. Program to check the given number is palindrome.
7. Program to find the roots of quadratic equation.
8. Program to perform the addition of Matrices
9. Program to demonstrate string functions.
10. Program to demonstrate pointers in C
11. Program to calculate the multiplication of two matrices.
12. Program to demonstrate the Employee database using structure.

1.7 Introduction to Linux

Brief History of linux:

What is Unix?, Unix Components, Using Unix, Commands in Unix, Some Basic Commands–echo, tput, who, uname, date, cal, cat, Getting Help – man. Files and File Organization: Unix Files,Categories of Files,Hidden Files, File System, Path Names, Home Directory, Directory Commands – pwd, cd, mkdir, rmdir, dot and double dot, File Related Commands
– cat, ls, mv, rm, wc and split, Wild Cards, Displaying the Contents of a File – more, head, tail, Comparing Files – cmp, diff and comm.

File Attributes and Permissions:

Ownership of Files, File Attributes, File Command, Changing File Permission, Changing the Owner of a File, Changing the group of a File, umask Command. Standard I/O, RedirectionPipes and Filters: Standard I/O, Redirection, Pipe and Pipeline, Filter, tee command, sort command, uniq command and tr command.

The vi Editor:

vi Editor, Editing with vi, Moving the Cursor, Editing, Copying and Moving Text, Pattern Searching, Repeating the Last Editor Command, Undoing Commands, Joining Lines,

1.8 Linux Lab

1. Write shell script to add two numbers.
2. Write shell script to find maximum of two numbers.
3. Write a Shell script to find biggest of three numbers.
4. Write a shell script to exchange the contents of two number
5. Write a Shell script that accepts a filename, starting and ending line numbers asarguments and displays all the lines between the given line numbers
6. Create A File Called Wlcc.Txt With Some Lines And Display How Many Lines, Words And Characters Are Present In That File.

PART – B

1. Write a shell script to change date format. Show the time taken in execution of this script
2. Write a shell script to count lines, words & characters in its input. (do not use wc).
3. Write Shell Script To Explain "If" Command.
4. Write Shell Script To Swap Two Numbers.
5. Write Shell Script To Illustrate For And While Loop.
6. Write a shell script to compute GCD & LCM of two numbers.
7. Write a shell script to find whether a given number is prime

1.9 OEC-1

Introduction to Banking

Definition and Meaning of Banking, history, features and evolution of Banking in India, functions and types of banking, Importance of Banking in the Economy.

Banking Operations and Services

Basic Banking Operations, Types of Bank Accounts, procedures for opening bank accounts Banking, Banker-Customer Relationship, importance and procedures of Know Your Customer (KYC).

Negotiable Instruments

Meaning, definition, Characteristics and kinds of Negotiable Instruments. Meaning, definition and features of promissory notes, bills of exchange and cheque. Difference between promissory note and bill of exchange. Parties to Negotiable Instruments, cheque – meaning and types of cheque.

Recent Developments in Banking

New technology in banking - E-services, Debit card and Credit cards, internet banking, ATM, Electric Fund Transfer, MICR, RTGS, NEFT, E-banking, core banking, and mobile banking - their advantages and disadvantages, Difference between Traditional banking and Online banking.

2.1 KAANNDA SHRAVANA-2

ಕನ್ನಡ ಭಾಷಾ ಪತ್ಯ : ಕನ್ನಡ ಶ್ರಾವಣ–2

ಘಟಕ–1 : ವೈಜ್ಞಾನಿಕ ಮನೋಧರ್ಮ

1. ವೈಜ್ಞಾನಿಕ ಮನೋಧರ್ಮ : ಜಿ.ಟಿ. ನಾರಾಯಣರಾವ್
2. ನಮ್ಮ ಅಳತೆಯನ್ನು ಮೀರಲಾರದ ದೇವರು : ಶಿವರಾಮ ಕಾರಂತ
3. ಇರುವುದೊಂದೇ ಭೂಮಿ : ನಾಗೇಶ ಹೆಗಡೆ
4. ಮಾನವತಾವಾದ : ಬಿ.ವಿ. ವೀರಭದ್ರಪ್ಪ

ಘಟಕ–2 : ಸಂಕೀರ್ಣ : ವಿಜ್ಞಾನ ಮತ್ತು ಸಾಹಿತ್ಯ ಸಂಬಂಧ / ಮಹಿಳಾ ಸಂವೇದನೆ

1. ವೈಜ್ಞಾನಿಕ ದೃಷ್ಟಿ : ಕೆ.ವಿ. ನಾರಾಯಣ

1. ವಿಜ್ಞಾನದ ನೈತಿಕ ಮುಖಗಳು : ಬಿ.ಸಿ. ರಾಯ್
2. ಕಾಯಕ : ಅಲಕ್ಷಿತ ವಚನಕಾರ್ತಿಯರ ವಚನಗಳು
3. ಒಮ್ಮೆ ಹೆಣ್ಣಾಗು ಪ್ರಭುವೇ : ಬಾನು ಮುಸ್ತಾಕ್

2.1 KATHA SOURABHA

<u>गद्य:</u>

1) एक टरकरी भर ममट्टी- मामोव राव सप्रे

2) भरराम का जीव- 5रर्शरोंकर परसाई

3) चीफ़ की दावत- भीष्म सा5नी

कफ़न- प्रेमचरोंद पद्य:

<u>पद्य:</u>

1) स्वयमागत – म ममथीशरण गुप्त
2) आओ 5म पथ से 5ट जाए – र्◌ो◌ो◌ो◌ा◌ो◌ो◌ो◌ा◌ो◌ो◌ो◌ा◌ो◌ा◌ो . 5रर्वरोंशराय बच्चन
3) सामोंी – कबीरदासेक दर5◌ो◌ो◌ो◌े
4) पद - मीराबाई

2.2 ENGLISH INSIGHT II

PROSE

1. The Thief's Story - Ruskin Bond
2. Function of Education -J Krishnamurthy
3. Daughter - Lata Jagtiani
4. Wings of fire - Dr A.P.J, Abdul Kalam

POETRY

1. Once Upon a Time -Gabriel Okara
2. The Man He Killed -Thomas Hardy
3. If - Rudyard Kipling
4. Still I Rise -Maya Angelou

LANGUAGE PRACTICE

1. Question Forms
2. Active Voice and Passive Voice
3. Letter Writing
4. Content Writing
5. Story Writing
6. Comprehension Reading Poetry

2.3 Data structure using C

Introduction to data structures:

Definition; Types of data structures - Primitive & Non-primitive,Linear andNon-linear; Operationsondatastructures. Dynamic memory allocation: Static & Dynamic memory allocation; Memory allocation and de-allocation functions–malloc, calloc, realloc and free. Algorithm Specification, Performance Analysis, Performance Measurement Recursion: Definition ; Types o recursions ; Recursion Technique Examples - GCD, Binomial coefficient $^{n}C_{r}$, Towers of Hanoi; Comparison between iterative and recursivefunctions.

Arrays:

Basic Concepts–Definition, Declaration, Initialization, Operations on arrays;Types of arrays; Arrays as abstract data types (ADT); Representation of Linear Arrays in Memory Traversing linear arrays; Inserting and deleting elements; Sorting – Selection Sort, Bubble sort, Quick sort, Selection sort, Insertion sort; Searching-Sequential Search Binary Search;Iterative and Recursive searching.

Stacks: Basic Concepts – Definition and Representation of stacks; Operations on stacks; Applications of stacks; Infix, postfix and prefix notations; Conversion from Infix to postfix using stacks; Evaluation of post fix expression using stacks.

Queues and Linked list:

Queues: Basic Concepts–Definition and Representation of queues; Types of queues Simple queues, Circular queues, Double ended queues, Priority queues;Operations of Simple queues.

Linked list: Basic Concepts – Definition and Representation of linked list, Types of linked lists - Singly linked list, Doubly liked list, Header liked list, Circular linked list; Representation of Linked list in Memory Operations on Singly linked lists–Traversing, Searching, Insertion, Deletion; Memory allocation; Garbage collection

Trees:

Trees: Definition; tree terminologies-node, root node, parent node, ancestors of a node Siblings, terminal, and non terminal nodes, degree of node, level, edge, path, depths.Binary tree: type of binary trees – strict binary tree, complete binary tree, binary search tree and heap tree; Array representation of binary tree. Traversal of binary tree; preorder, in order and post order traversal

2.4 Data Structures using C Lab()

- Program to find GCD using recursive function
- Program to display Pascal Triangle using binomial function
- Program to generate n Fibonacci numbers using recursive function
- Program to implement Towers of Hanoi
- Program to implement dynamic array, find smallest and largest element of the array
- Program to create two files to store even and odd numbers.
- Program to create a file to store student records.
- Program to read the names of cities and arrange them alphabetically
- Program to sort the given list using selection sort technique.
- Program to sort the given list using bubble sort technique.

PART – B

- Program to sort the given list using insertion sort technique
- Program to sort the given list using quick sort technique.
- Program to sort the given list using merge sort technique
- Program to search an element using linear search technique
- Program to search an element using recursive binary search technique
- Program to implement Stack
- Program to convert an infix expression to postfix
- Program to implement simple queue
- Program to implement linear linked list
- Program to display traversal of a tree.

2.5 Object Oriented Concepts using JAVA()

Introduction to Java: Basics of Java programming, Data types, Variables, Operators, Control structures including selection, Looping, Java methods, Overloading, Math class, Arrays in java.

Objects and Classes: Basics of objects and classes in java, Constructors, Finalizer, Visibility modifiers, Methods and objects, Inbuilt classes like String, Character, String Buffer, File, this reference.

Inheritance and Polymorphism: Inheritance in java, Super and subclass, Overriding, Object class, Polymorphism, Dynamic binding, Generic programming, Casting objects, Instance of operator, Abstract class, Interface in java, Package in java, UTIL package.

Multithreading in java: Thread life cycle and methods, Runnable interface, Thread synchronization, Exception handling with try catch-finally, Collections in java, Introduction to JavaBeans and Network Programming.

Event and GUI programming: Event handling in java, Event types, Mouse and key events, GUI Basics, Panels, Frames, Layout Managers: Flow Layout, Border Layout, Grid Layout, GUI components like Buttons, Check Boxes, Radio Buttons, Labels, Text Fields, Text Areas, Combo Boxes, Lists, Scroll Bars, Sliders, Windows, Menus, Dialog Box, Introduction to swing, Exceptional handling mechanism.

I/O programming: Text and Binary I/O, Binary I/O classes, Object I/O, Random Access Files.

2.6 Java Lab

1. Write a java program to find the Fibonacci series using recursive and non- recursive functions.
2. Write a Java program to read the radius of a circle and to find the area and circumference.
3. Write a java program for Method over loading and Constructor over loading.
4. Write a Java program to find N prime numbers reading N as command line argument.
5. Write a program to find factorial of N numbers reading N as command line argument.
6. Write a java program that checks whether a given string is palindrome or not
7. Write a program to read N numbers and sort them using one-dimensional arrays.

PART – B

1. Write a Java program to illustrate Method overloading.
2. Write a Java program demonstrate Single Inheritance.
3. Write a program to illustrate Constructor Overloading.
4. Write a program to illustrate Method Overriding.
5. Write a Java program to demonstrating Multithreading.
6. Write a Java program to demonstrate Exception Handling.
7. Write an Applet program to display Geometrical Figures usingobjects.
8. Write an Applet program to change the background color randomly.
9. Write an Applet program to change the color of applet using combo box.
10. Write an Applet program to implement Mouse events.

2.7 OEC-2

Course Content

Managing Workforce

Workforce Management

Workforce Management – Origin, Meaning, Definition, importance, advantages. HR Manager – meaning, qualities, qualifications, roles and responsibilities. HR Department – meaning, duties of department, hierarchy in the department.

Resume Building and Recruitment

Resume- Meaning, components, importance, format; Curriculum vitae – meaning, components, importance, format; Bio data: meaning, components, importance, format. Recruitment – meaning, definition, importance, methods of recruitment, types of recruitment – meaning and importance of AI Powered recruitment, Social media and digital recruitment, Virtual, Gamification, hackathons, employee referrals.

Wage and Salary administration

Wage – meaning, methods of wage payment, advantage, disadvantage of each method, Salary – meaning, definition, components of salary, Cost to company- meaning, take home/Net salary- meaning. Preparing a sample salary format.

Industrial Dispute

Industrial dispute – meaning, definition, parties involved in industrial dispute, strike – meaning, definition, types of strikes, lock down- meaning, definition, layover- meaning, definition. Case studies: Parry and Co Ltd vs P.C. Pal 1970; Workmen of Hindustan Lever Ltd vs Hindustan Lever Ltd (1999), Ajaypal Singh vs Haryana Warehousing Corporation (2014), Navbharat Press Karmachari Kalyan Sangh VS. State of Chattisgarh (2015), State of U.P. & Others vs P.O. Labour Court & Another (2015), Sunil Kumar vs

N.C.C.F. (2022), Motherson Automotive Technologies &... vs The Deputy Commissioner of Labour (2024) etc.

3.1 Data Base System Concepts

Course Content

Introduction: application, database system vs. file system, view of data models- Hierarchical, Relational, database languages, database users and administrators, transaction management, Database system structure, application architecture.E_R ModelBasic-concepts, constraints, keys, Design Issues, E-R diagram, weak entity sets, Extended E-R features, Design of an ER database schema.

Relational Model: Structure of Relational Databases, Relational Algebra, Extended Relational Algebra Operations, Modification of the database, Views. SQL Background, Basic structure, set operation, aggregate functions, NULL values, subqueries, views, Modification of the Database, joined Relations, DDL.

Normalization: Introduction, 1NF, 2NF and FDs, 3NF and BCNF, 4NF and MVDs, 5NF and PJNF, Closure of a set of FDs, MVDs, and Loss less join.

Transaction Management: Introduction, Transaction concepts and properties, States, concurrent execution, Scheduling of Transactions, Operations Conflicts, Testing for serializability. Concurrency Control Techniques**:** Introduction, Lock-based Protocols, Time stamp Protocols, Failure Classification.Recovery and Atomicity Introduction, Categorization of Recovery Algorithms.

3.2 Data Base System Lab

Course Content

1. Implementation of DDL commands of SQL with suitable examples
   - Create table
   - Alter table
   - Drop Table
2. Implementation of DML commands of SQL with suitable examples
   - Insert
   - Update
   - Delete
3. Implementation of different types of function with suitable examples
   - Number function
   - Aggregate Function
   - Character Function
   - Conversion Function
   - Date Function
4. Implementation of different types of operators in SQL
   - Arithmetic Operators
   - Logical Operators
   - Comparison Operator
   - Special Operator
   - Set Operation
5. Implementation of different types of Joins
   - Inner Join
   - Outer Join
   - Natural Join etc.
6. Study and Implementation of
   - Group By & having clause
   - Order by clause
   - Indexing
7. Study & Implementation of
   - Sub queries
   - Views
8. Study & Implementation of different types of constraints.
9. Study & Implementation of Database Backup & Recovery commands. Study & Implementation of Rollback, Commit, Savepoint.
10. Creating Database /Table Space Managing Users: Create User, Delete User Managing roles: -Grant, Revoke.

3.3 Python Programming
Course Content

Introduction: Features and applications of python, python versions installation of python, python command line mode, simple python program. Python Basics Identifiers; Keywords; Statements and Expressions Variables; Operators; Precedence and Association; Data Types; Indentation Comments; Built-in Functions- Console Input and Console Output, Type Conversions; Python Libraries; Importing Libraries with Examples.

Python Control Flow: Types of Control Flow; Control Flow Statements- if else, elif, while loop, break, continue statements, for loop Statement; range() andexit()functions.Exception Handling Types of Errors; Exceptions; Exception Handling usingtry, except and finally. Python Functions: Types of Functions; Function Definition Syntax Function Calling, Passing Parameters/arguments, the return statement; Default Parameters; Command line Arguments; Key Word Arguments; Recursive Functions; Scope and Lifetime of Variables in Functions.

Strings: Creating and Storing Strings; Accessing Sting Characters; the str function; Operations on Strings Concatenation, Comparison, Slicing and Joining, Traversing; Format Specifiers; Escape Sequences; Raw and Unicode Strings; Python String Methods.Lists Creating Lists; Operations on Lists; Built-in Functions on Lists Implementation of Stacks and Queues using Lists; Nested Lists. Dictionaries Creating Dictionaries; Operations on Dictionaries; Built-in Functions on Dictionaries; Dictionary Methods; Populating and Traversing Dictionaries.File Handling File Types; Operations on Files Create, Open, Read, and Write, Close Files; File Names and Paths; Format Operator.

Object Oriented Programming: Classes and Objects; Creating Classes and Objects; Constructor Method; Classes with Multiple Objects; Objects as Arguments; Objects as Return Values; Inheritance- Single and Multiple Inheritance, Multilevel and Multipath Inheritance; Encapsulation- Definition, Private Instance Variables; Polymorphism- Definition, Operator OverloadingGU Interface The tkinter Module; Window and Widgets; Layout Management- pack, grid and place.

Python SQLite: The SQLite3 module; SQLite Methods- connect, cursor execute, close; Connect to Database; Create Table; Operations on Tables- Insert, Select, Update. Delete and Drop Records. Data Analysis NumPy- Introduction to NumPy, Array Creation using NumPy, Operations on Arrays; Pandas- Introduction to Pandas, Series and Data Frames, Creating Data Frames from Excel Sheet and .csv file, Dictionary and Tuples. Operations on Data Frames.Data Visualisation Introduction to Data Visualisation; Matplotlib Library Different Types of Charts using Pyplot- Line chart, Bar chart and Histogram and Pie chart.

3.4 Python Lab

Course Content

PART – A

1. Write a program to demonstrate different number data types in python
2. Write a program to perform different arithmetic operations on numbers inpython
3. Write a program to create, concatenate and print a string and accessing sub-string from given string
4. Write a python script to print the current date in the fallowing format "Sun May29 02:26:23 IST 2017"
5. Write a program to create, append and remove lists in python
6. Write a program to demonstrate working with tuples in python
7. Write a program to demonstrate working with dictionaries in python
8. Program to find the largest number among the three input numbers
9. Program to convert temperature in Celsius to Fahrenheit
10. Program to create a calculator program

PART – B

1. Write a python program to construct the following pattern, using a nested forloop
2. Write a python program to find the factorial of a number using recursion
3. Write a python program to define a module to find Fibonacci numbers and import the module to another program
4. Program to demonstrate usage of basic regular expression
5. Program to create a GUI using Tkinter module
6. Program to demonstrate Exceptions in Python
7. Program to Drawing Line chart and Bar chart using Matplotlib
8. Program to Drawing Histogram and Pie chart using Matplotlib
9. Create Array using NumPy and Perform Operations on Array
10. Demonstrate in order, pre order and post order traversal.

3.5Operating System
Course Content

Introduction to operating systems: System structures,what operating systems do; Computer System organization; Computer System architecture; Operating System structure; Operating System operations; Process management; Memory management; Storage management; Protection and Security; Distributed system; Special-purpose systems; Computing environments.

Operating System Services: User - Operating System interface; System calls; Types of system calls; System programs; Operating system design and implementation; OperatingSystem structure; Virtual machines; Operating System debugging, Operating Systemgeneration; System boot.

Process Management: Process concept; Process scheduling; Operations on processes; Inter process communication Multi-threaded Programming: Overview; Multithreading models; Thread Libraries; Threading issues. Process Scheduling: Basic concepts; Scheduling Criteria; Scheduling Algorithms; Thread scheduling; Multiple-processor scheduling.

Process Synchronization: Synchronization: The critical section problem; Peterson's solution,Synchronization hardware; Semaphores; Classical problems of synchronizationDeadlocksSystem model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock.

Memory Management: Memory management strategies, Background, Swapping, Contiguous memory allocation, Paging-Structure of page table, Segmentation. Virtual Memory Management: Background, Demand paging, Copy-on-write, Page replacement; Allocation of frames; Thrashing.

3.6Discrete Mathematics

Course Content

The Foundation: Logic and Proofs: Propositional Logic, Applications of Propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference, Introduction to Proofs, Proof Methods and Strategy.

Basic Structures**:** Sets, Set Operations, Functions, Sequences, Sums and Matrices.

Counting: Basics of Counting, Pigeonhole Principle, Permutation and Combination, Binomial Coefficient and Combination, Generating Permutation and Combination.

Induction and Recursion: Mathematical Induction, Strong Induction and Well – Ordering, Recursive Definitions and Structural Induction, Recursive Algorithms, Program Corrections.

Relation: Properties of Relation, Composition of Relation, Closer Operation on Relation, Equivalence Relation and Partition. Operation on Relation, Representing Relation.

Graphs: Graphs and Graph Models, Graph Terminology and Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Shortest-Path Problems, Planar Graphs, Graph Coloring.

3.7 OEC-3

Course Content

Introduction to E-Commerce and Technology Infrastructure:
Working of Web-HTML Markup for Structure Creating simple page-Marking up text- Adding Links-Adding Images- Table Markup - Forms-HTML5.

Building an E-Commerce Website, Mobile Site and Apps: Systematic approach to build an E-Commerce: Planning, System Analysis, System Design, Building the system, Testing the system, Implementation and Maintenance, Optimize Web Performance– Choosin gb\ hardware and software–Other E-Commerce Site tools–Developing a Mobile Website and Mobile App.

E-Commerce Security and Payment Systems: E-Commerce Security Environment– Security threats in E-Commerce–Technology Solutions: Encryption, Securing Channels of Communication, Protecting Networks, Protecting Servers and Clients – ManagementPolicies, Business Procedure and Public Laws-Payment Systems.

Business Concepts in E-Commerce:
Digital Commerce Marketing and Advertising strategies and tools– Ethical, Social, Political Issues in E-Commerce.

Project Case Study: Identify Key components, strategy, B2B, B2C Models of E- commerce Business model of any e-commerce website - Mini Project: Develop E- Commerce project in any one of Platforms like Woo-Commerce, Magento or Open cart. Internet Marketing Technologies– Social Marketing–Mobile Marketing– Location based Marketing –

4.1 Introduction to AI

Introduction to Artificial Intelligence:

AI: Definition, History and evolution of AI, Applications of AI in various domains; Intelligent agents; Environments, Agent based programs, Structure of agents; Future of AI: Emerging trends in AI, AI's role in automation and Industry;

Intelligent Agents:

Typical Intelligent Agents: Reflex agents and their implementation, Model-based and goal-based agents, Utility-based agents and Goal based agents; AI Problem Solving: Definition, Problem Representation (State-Space Model), Components of a Problem (Initial State, Goal State, Operators);

Search Algorithms and Strategies:

Uninformed Search: Characteristics, Types (BFS, DFS, UCS & DLS); Informed Search (Heuristic Search): Characteristics, Greedy Best-First Search, A* search and AO* search, Search Strategies;

Logic and Knowledge-Based Systems in AI:

First-Order Predicate Logic (FOPL): Introduction, Syntax and Semantics, Representation; Prolog Programming Unification: Introduction, Syntax and Semantics, Representation; Forward Chaining-Backward Chaining, Knowledge Representation (Knowledge based agents, Wumpus world)

Machine Learning and AI Applications:

Learning: Forms of learning, supervised learning, Machine learning; Decision trees,
ANN, SVM; NLP; Text classification and information retrieval, speech recognition; Computer vision and Robotics;

4.2 Artificial Intelligence Lab

Course Content

1. Write a Python program to Implement BFS
2. Write a Python program to Implement DFS
3. Write a Python program implement Water Jug Problem.
4. Write a Python program to Implement Tic-Tac-Toe Game.
5. Write a Python program to Implement A* Algorithm
6. Write a Python program to Use Python Libraries (math, NumPy, SciPy)
7. Write a Python program to Compute Statistical Metrics.
8. Write a Python program to Implement S Algorithm
9. Write a Python program to Implement Linear Regression.
10. Write a Python program to Use Pandas and Matplotlib for ML
11. Design a Python program using Matplotlib to visualize the relationship between two variables (e.g., age and income) in a scatter plot and customize it with labels, titles, and gridlines.
12. Create a Python program using Pandas to manipulate a dataset by filtering rows with specific criteria, calculating the mean of a column, and saving the results to a new CSV file.

4.3 Design and Analysis of Algorithm Using C

Course Content

Introduction to Algorithms

Fundamentals of Algorithmic problem solving, Fundamentals of the Analysis of Algorithm Efficiency, Analysis Framework, Measuring the input size, Units for measuring Running time, Orders of Growth, Worst-case, Best- case and Average-case efficiencies., Asymptotic Notations and Basic Efficiency classes, Mathematical Analysis of Recursive and Non recursive algorithms.

Brute Force & Exhaustive Search

Introduction to Brute Force approach, Selection Sort and Bubble Sort, Sequential search, Exhaustive Search- Travelling Salesman Problem and Knapsack Problem, Depth First Search (DFS), Breadth First Search(BFS)

Decrease-and-Conquer

Introduction, Insertion Sort, Topological Sorting Divide- and- Conquer: Introduction, Merge Sort, Quick Sort, Binary Search, Binary Tree traversals and related properties.

Introduction, Optimal storage on Tapes, Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm ,Fractional Knapsack problem, Lower-Bound Arguments, Decision Trees, P Problems, NP Problems, NP-Complete Problems, Challenges of Numerical Algorithms.

BackTracking:

General method, Applications- n-queen problem, Sum of subsets problem, Graph coloring and Hamiltonian cycles.

4.4 Design and Analysis of Algorithm Lab
Course Content

Part A

1. Write a program to perform Linear Search on array elements.
2. Write a program to find factorial of number using recursion.
3. Write a program to find Fibonacci of a number using recursion.
4. Write a program to sort the elements using Selection sort.
5. Write a program to sort the elements using Bubble sort.
6. Write a program to perform Graph traversal using Depth First Search.
7. Write a program to perform Graph traversal using Breadth First Search.
8. Write a program to implement Binary Search using Divide and Conquer technique.

Part B

1. Write a program to implement Knapsack problem using Greedy Technique.
2. Write a program to perform Merge sort on array elements.
3. Write a progam to sort the array elements using Quick sort .
4. Write a program to estimate a path of a graph using Prim's algorithm.
5. Write a program to estimate a path of a graph using Kruskal's algorithm.

4.5 Computer Network

Course Content

Introduction to Computer Networks

The fundamental concepts of computer networks, their uses, and design principles. network hardware and software, reference models such as the OSI and TCP/IP, an overview of data communication fundamentals, circuit switching, packet switching, and a brief history of the evolution of networking technologies.

The Physical Layer

Physical aspects of data transmission, transmission media (guided and unguided), signal encoding, modulation techniques, data link interfaces, bandwidth, latency, noise, and error detection techniques, multiplexing methods (TDM, FDM, WDM) and the role of network hardware like repeaters, hubs, and modems.

The Data Link Layer and Medium Access Control

The data link layer's responsibilities, such as framing, error detection, correction, stop-and-wait ARQ, go-back-N, selective repeat, Medium Access Control (MAC) is explored with a focus on Ethernet, wireless LANs, and other multiple access techniques, ALOHA, CSMA/CD, and CSMA/CA.

The Network Layer

Routing algorithms such as distance vector, link state, and hierarchical routing are discussed. Quality of Service (QoS), congestion control, and virtual circuit switching.

The Transport and Application Layers

The transport layer and application layer protocols. Topics include TCP, UDP, and their respective functionalities like flow control, error handling, and connection management. application protocols such as HTTP, FTP, SMTP, DNS, and SNMP. network security basics, and cryptographic protocols (SSL/TLS).

4.6 Cyber security

Course Content

Introduction to Cyber security: Defining Cyberspace and Overview of Computer and Web-technology, Architecture of cyberspace, Communication and web technology, Internet, World wide web, Advent of internet, Internet infrastructure for data transfer and governance, Internet society, Regulation of cyberspace, Concept of cyber security, Issues and challenges of cyber security.

Cyber-crime and Cyber law: Classification of cybercrimes, Common cyber crimes- cyber-crime targeting computers and mobiles, cyber crime against women and children, financial frauds, social engineering attacks, malware and ransomware attacks, zero day and zero click attacks, Cybercriminals modus-operandi, Reporting of cyber crimes, Remedial and mitigation measures, Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber-crime and offences, Organisations dealing with Cybercrime and Cyber security in India, Case studies.

Social Media Overview and Security: Introduction to Social networks. Types of Socialmedia, Social media platforms, Social media monitoring, Hash tag, Viral content, Social media marketing, Social media privacy, Challenges, opportunities and pitfalls in online social network, Security issues related to social media, Flagging and reporting of inappropriate content, Laws regarding posting of inappropriate content, Best practices for the use of Social media, Case studies.Definition of E- Commerce, Main components of E- Commerce, Elements of E-Commerce security, E-Commerce threats, E-Commerce security best practices, Advantage of e-commerce, Survey of popular e-commerce sites.

Introduction to digital payments: Components of digital payment and stakeholders, Modes of digital payments- Banking Cards, Unified Payment Interface (UPI), e-Wallets, Unstructured Supplementary Service Data (USSD), Aadhar enabled payments, Digital payments related common frauds and preventive measures. RBI guidelines on digital payments and customer protection in unauthorized banking transactions. Relevant provisions of Payment Settlement Act,2007.

4.7 Network Security

Course Content

Introduction - Attacks, Services, and Mechanisms – Security Attacks: Passive Attacks – Active Attacks – Security Services – Model for Internetwork Security – Internet Standards and RFCs: Internet Society – RFC Publication – Standardization Process – Non-Standardization Track Documents – Conventional Encryption and Message Confidentiality: Conventional Encryption Principles: Cryptography – Cryptanalysis – Feistel Cipher Structure – Conventional Encryption Algorithms – Cipher Blocks Models of Operation: Cipher Block Chaining Mode – Cipher Feedback Model – Location of Encryption Devices – **Key distribution.**

Public Key Cryptography and Message Authentication - Approaches to Message Authentication: Authentication using Conventional Encryption – Message Authentication without Message Encryption – Message Authentication Code – One Way Hash Function – Secure Hash Function and HMAC: Hash Function Requirements – Simple Hash Functions the SHA–1 Secure Hash Function – Other Secure Hash Functions – HMAC Design Objectives – HMAC Algorithm – Public Key Cryptography Principles: Public Key Encryption Structure Application for Public Key Cryptosystems – Requirements for Public **Key Cryptography** – Public Key Cryptography Algorithms: RSA Public Key Encryption Algorithms – Diffie-Hellman Key Exchange – Other Public Key Cryptography Algorithms – Digital Signature – Key Management.

Electronic Mail Security - Pretty Good Privacy (PGP): Notation – Operational Description – Cryptography Keys and Key Rings – Public Key Management – S/MIME: RFC 822 – Multipurpose Internet Mail Extensions – S/MIME Functionality – S/MIME Messages – S/MIME Certificate Processing – Enhanced Security Services – IP Security: IP Security **Overview**: Application of IPSec – Benefits of IPSec – Routing Applications – IP Security Architecture: IPSec Documents – IPSec Services – Security Associations – Transport and **Tunnel Modes** – Authentication Header – Encapsulating Security Payload – Combining Security Associations – Key Management: Oakley Key Determination Protocol – ISAKMP

Web security - Web security Requirements: Web security Threats – Web Traffic Security Approaches – Secure Socket Layer (SSL) and Transport Layer Security (TLS): SSL Architecture – SSL Record Protocol – Change Cipher Spec Protocol – Alert Protocol - Handshake Protocol – Cryptographic Computations – Transport Layer Security – Secure

Electronics Transaction (SET): SET Overview – Dual Signature – Payment Processing – Network Management Security: Basic Concepts of SNMP: Network Management Architecture – Network Management Protocol Architecture – Proxies – snmpv2 – snmpv1 Community Facility – snmpv3: SNMP Architecture – Message Processing and the User Security Model – View Based Access Control.

4.8 Cloud Computing

Course Content

Introduction

Cloud Computing Basics- What is Cloud Computing? History, Characteristic Features, Advantages and Disadvantages. Cloud Different Computing Paradigms- Parallel Computing, Distributed Computing, Cluster Computing, Grid Computing, Cloud Computing etc., Comparison of various Computing Technologies;

Virtualization

Introduction, Characteristics of Virtualized, Environments Taxonomy of Virtualization Techniques, Execution Virtualization, Other Types of Virtualization, Virtualization and Cloud Computing, Pros and Cons of Virtualization, Technology Examples

Cloud Computing Architecture

Cloud Reference Model:Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS); Types of Clouds:Private,public,hybrid clouds; Economics of the Cloud, Open Challenges.

Cloud Security

Risks, Top concern for cloud users, privacy impact assessment, trust, OS security, VM Security, Security Risks posed by shared images and management OS.

Cloud Platforms

Cloud Platforms in Industry Amazon web servicesCompute services, Storage services, Communication services, Additional services. Google AppEngine: - Architecture and core concepts, Application life cycle, Cost model, Observations.

Advanced Cloud Computing and Future Trends

Emerging Trends in Cloud Computing,Cloud Automation,Serverless Computing,Big Data and Cloud Integration,AI and Machine Learning in Cloud,Cloud Governance, Sustainability in Cloud.

5.1 Web Technology

Course Content

Fundamentals of Web: Fundamentals of Web: A Brief Introduction to the Internet, The World Wide Web, Web Browsers, Web Servers, Uniform Resource Locators,

Multipurpose Internet Mail Extensions, The Hypertext Transfer Protocol. Introduction to XHTML: Basic Syntax, Standard XHTML Document Structure, Basic Text Markup, Images, Hypertext Links, Lists, Tables, Forms, Syntactic Differences between HTML and XHTML.

CSS & Basics of JavaScript: CSS Levels of Style Sheets, Style Specification Formats, Selector Forms, Property Value Forms, Font Properties, List Properties, Color, Alignment of Text, The Box Model, Background Images, The and Tags, Conflict Resolution. The Basics of JavaScript: Overview of JavaScript, Object Orientation and JavaScript, General Syntactic Characteristics, Primitives Operations and Expressions

JavaScript and XHTML Documents JavaScript: Screen Output and Keyboard Input, Control Statements, Object Creation and Modification, Arrays, Functions, Pattern Matching Using Regular Expressions, Errors in Scripts.
XHTML Documents: The JavaScript Execution Environment, The Document Object Model, Element Accessing JavaScript, Events and Event Handling.

Event Handling & Dynamic Documents with Java Script Event Handling: Handling Events from Body Elements, Handling Events from Button Elements, Handling Events from Text Box and Password Elements, the DOM 2 Event Model. Dynamic Documents with JavaScript: Introduction, Positioning Elements, Moving Elements, Element Visibility, Changing Colors and Fonts, Dynamic Content, Stacking Elements, Locating the MouseCursor, reacting to a Mouse Click, Slow Movement of Elements.

5.2 Web Technology Lab

Course Content

PART – A

1. Create a web page to display the course syllabus.
2. Design student details in first page and result in next page, link both pages using hyper links.
3. Create a web page for class time table
4. Develop a web page with 2 or more images, move images in different directions and write hyper link to any one image.
5. Write a java script to design a simple calculator to perform the following operations sum, product, difference and quotient.
6. Write a java script that calculate the squares and cubes of the numbers from 0 to 10 and outputs HTML text that displays the resulting values inan HTML table format.
7. Design login page and validate it using JavaScript.
8. Develop a HTML Form which accepts two numbers. Write JavaScript code to execute arithmetic operations and displays the result.

PART – B

1. Create a form for Employee information. Write JavaScript code to findDA, HRA, PF, TAX, Gross pay, Deduction and Net pay.
2. Create a form for Student information. Write JavaScript code to find Total, Average, Result and Grade.
3. Develop web page with one text box and Write a JavaScript code to convertthe text entered in textbox toupper case.
4. Create a web page using two image files, which switch between one another as the mouse pointer movesover the images. Use the on Mouse Over and on Mouse Out.
5. Develop a HTML Form, which accepts any Mathematical expression. Write JavaScript code to Evaluates the expression and Displays the result.
6. Create a HTML form with 3 Textboxes. Write JavaScript code to validate input for numbers, alphabets, alphanumeric and verify that all 3 textboxes has been filled.
7. Write a user-defined function in JavaScript to find sum of N Number.
8. Write a JavaScript code to find factorial of a number using recursivefunction.

Course Content

5.3 C# and Dot Net Framework

Course Content

Introduction: Introduction to .Net, Two tier and Three tier client server model, .Net Architecture, Features of .Net, Advantages of .Net, .Net Framework, CLR, CTS, CLS, Assemblies, Memory management issues – Garbage Collector and collection process, Exception Handling, Code Access Security.

C# Languages Fundamentals: Need of c# ,C# pre-processor Directives , Features of C# , Creating a Simple c# Console Application, Identifiers and keywords , Data Types ,Variables and Constants Value type and reference types ,Boxing and un boxing types , Iterators, Constructs ,Control flow constructs Arrays , C# Enumerations.

Object- oriented Programming with C#: Namespaces, Class and objects, using this keyword, creating array of objects, defining partial Classes and Methods, Properties, Constructors and Destructors Static classes and static method, static variables, Encapsulations using properties, Inheritances, Interface, Exception Handling.

Graphical User interface with Windows Forms: Event Handling Control properties and layout labels, Textboxes, buttons, Group boxes and panels, checkboxes and radio button, Tooltips, List Box and Combo Box ,Group boxes Mouse-Even handling, Keyboard –Event Handling ,Understanding ADO.NET and Creating Connection string.

5.4 C# Lab

Course Content

PART – A

1. Develop a C#.NET console application to demonstrate the conditional statements.
2. Develop a C#.NET application to demonstrate the control statements.
3. Construct a console application to demonstrate the OOP concepts.
4. Develop programs to sorting a one dimensional array by accepting 5 elementsfrom user and arrange them in ascending order.
5. Develop a program to perform switch statements to display percentage of student.
6. Develop a program to perform single level inheritance in C#.NET.
7. Design an application to implement various string operations.
8. Demonstrate Multithreaded programming in C#.NET.
9. Write a program in C#.NET to find the sum of each row of given jagged array of 3 inner arrays.

PART – B

1. Develop an application in C#.NET that demonstrates the windows controls.
2. Write a program to perform Timer control properties Start (), Stop().
3. Develop an MDI application for Employee Pay-roll transactions inC#.NET.
4. Design an application to create a login form and validate it using message box.
5. Design an application to simulate the working of a font dialog box using combobox.
6. Design a Student Registration Application to store the student data in the databaseusing ADO.NET

5.5 Software Engineering

Course Content

The Product and The Process: Evolving role of software, software characteristics and components, Crisis, Software Myths, Software Engineering-A Layered Technology, Software process, linear sequential model, Prototyping model, RAD model, Evolutionary software process model. Project Management Concepts: The Management Spectrum, The People, The Product, The Process, and The Project. W5HH Principle. Software Process and Project Metrics: Measures, Metric Indicators, Metric in process and the Project Domains, Software Measurement, Metrics for software quality.

Software Project Planning: Project Planning Objectives, Software Project Estimation, decomposition Techniques, Empirical Estimation Models. Risk Analysis and Management: Software Risks, Risk Identification, Risk Projection, Risk Refinement and Risk Mitigation, Monitoring, and Management.

Analysis Concepts and Principles: Requirement analysis, communication techniques, analysis principles, software prototyping and specification. Analysis Modeling: Elements of analysis model, data modeling, functional modeling, behavioral modeling, the mechanics of structured analysis, data dictionary, otherclassical analysis methods.

Design Concepts and Principles: Software design and software engineering design process, design principles, design concepts, design methods, data design, architectural design and process,transform and transaction mappings, design post processing, architectural design optimization, interface design, procedural design.

Software Testing Techniques and Strategies: Fundamentals, Test case design, White box testing, Basis path testing, Control structure testing, Black box testing, Software testing strategies.

Software Configuration Management: Configuration management, maintenance costs, maintenance side effects, maintenance tissues. Software Quality Assurance: Quality Concepts.

5.6 Artificial Intelligence and Machine Learning
Course Content

Introduction to Artificial Intelligence

Overview of AI and its historical development Key challenges and goals in AI Ethical considerations in AI. Intelligent Agents: Definition and characteristics of intelligent agents Types of agents: simple reflex agents, model-based reflex agents, goal-based agents, utility-based agents

Problem Solving

Search algorithms: breadth-first search, depth-first search, A* search Informed search algorithms: heuristics, hill climbing, simulated annealing Knowledge and Reasoning: Propositional and first-order logic Inference in first-order logic Knowledge representation: semantic networks, frames, scripts

Learning

Machine learning basics Supervised and unsupervised learning Reinforcement learning. Planning: Definition and types of planning problems Classical planning approaches Planning with uncertainty

Special Topics and Applications

Overview of advanced AI topics (e.g., robotics, expert systems) Case studies and applications in real-world scenarios Future directions in AI research.

5.7 Artificial Neural Network

Course Content

Introduction

A Neural Network, Human Brain, Models of a Neuron, Neural Networks viewed as Directed Graphs, Network Architectures, Knowledge Representation, Artificial Intelligence and Neural Networks.

Learning Process

Error Correction Learning, Memory Based Learning, Hebbian Learning, Competitive, Boltzmann Learning, Credit Assignment Problem, Memory, Adaption, Statistical Nature of the Learning Process.

Single Layer Perceptrons

Adaptive Filtering Problem, Unconstrained Organization Techniques, Linear Least Square Filters, Least Mean Square Algorithm, Learning Curves, Learning Rate Annealing Techniques, Perceptron –Convergence Theorem, Relation Between Perceptron and Bayes Classifier for a Gaussian Environment.

Multilayer Perceptron

Back Propagation Algorithm XOR Problem, Heuristics, Output Representation and Decision Rule, Computer Experiment, Feature Detection. Back Propagation: Back Propagation and Differentiation, Hessian Matrix,

Generalization, Cross Validation, Network Pruning Techniques, Virtues and Limitations of Back Propagation Learning, Accelerated Convergence, Supervised Learning.

5.8 Big Data Analytics

Course Content

Getting an Overview of Big Data

What is Big Data? Types of Data, Elements of Big Data, Careers in Big data. Introducing Technologies for Handling Big Data: Distributed and Parallel Computing for Big Data, Cloud Computing and Big Data, and In-Memory Computing Technology for Big Data.

The Hadoop Distributed Filesystem

Design of HDFS,HDFS Concepts, Blocks, Namenodes and Datanodes, Block Caching, HDFS Federation ,HDFS High Availability , Hadoop file system, Data Flow: Anatomy of a File Read , Anatomy of a File Write, Coherency Model, Parallel Copying with distcp, Keeping an HDFS Cluster Balanced

YARN

Anatomy of a YARN Application Run, Resource Requests, Application Lifespan, Building YARN Applications, YARN Compared to MapReduce 1, Scheduling in YARN, Scheduler Options, Capacity Scheduler Configuration, Fair SchedulerConfiguration, Delay Scheduling, Dominant Resource Fairness.

Spark: Introduction to Spark, Difference between Hadoop and Spark. How MapReduce Works: Anatomy of a MapReduce Job Run: Job Submission, Job initialization,Task Assignment, Task Execution, Progress and Status Updates, Job Completion. Failures: Task Failure, Application Master Failure, Node Manager Failure, Resource Manager Failure. Shuffle and Sort: The Map Side, The Reduce Side,Configuration Tuning.

Task Execution

The Task Execution Environment, Speculative Execution, Output

Committers MapReduce Types and Formats: The Default MapReduce Job, Input Formats: Input Splits and Records, Text Input, Binary Input, Multiple Inputs, Database Input (and Output). Output Formats: Text Output, Binary Output

6.1 Full Stack Development with Integrated Lab

Course Content

MVC based Web Designing

Web framework, MVC Design Pattern, Django Evolution, Views, Mapping URL to Views, Working of Django URL Confs and Loose Coupling, Errors in Django, Wild Card patterns in URLS.

Django Templates and Models

Template System Basics, Using Django Template System, Basic Template Tags and Filters, MVT Development Pattern, Template Loading, Template Inheritance, MVT Development Pattern. Configuring Databases, Defining and Implementing Models, Basic Data Access, Adding Model String Representations, Inserting/Updating data, Selecting and deleting objects, Schema Evolution.

Django Admin Interfaces and Model Forms.

Activating Admin Interfaces, Using Admin Interfaces, Customizing Admin Interfaces, Reasons to use Admin Interfaces. Form Processing, Creating Feedback forms, Form submissions, custom validation, creating Model Forms, URLConf Ticks, Including Other URLConfs. Generic Views and Django State Persistence: Using Generic Views, Generic Views of Objects, Extending Generic Views of objects, Extending Generic Views. MIME Types, Generating Non- HTML contents like CSV and PDF, Syndication Feed Framework, Sitemap framework, Cookies, Sessions, Users and Authentication.

jQuery and AJAX Integration in Django

Ajax Solution, Java Script, XHTML HttpRequest and Response, HTML, CSS,JSON, iFrames, Settings of Java Script in Django, jQuery and Basic AJAX, jQuery AJAX Facilities, Using jQuery UI Autocomplete in Django.

- Laboratory Component:
- ➢ Installation of Python, Django and Visual Studio code editors can be demonstrated.
- ➢ Creation of virtual environment, Django project and App should be demonstrated

- Develop a Django app that displays current date and time in server
- Develop a Django app that displays date and time four hours ahead and four hours before as an offset of current date and time in server.
- Develop a simple Django app that displays an unordered list of fruits and ordered list of selected students for an event.
- Develop a layout.html with a suitable header (containing navigation menu) and footer with copyright and developer information. Inherit thislayout.html and create 3 additional pages: contact us, About Us and Home page of any website.
- Develop a Django app that performs student registration to a course. It should also display list of students registered for any selected course. Create students and course as models with enrolment as Many To Many field.
- Develop a Model form for student that contains his topic chosen forproject, languages used and duration with a model called project.
- Create a generic class view which displays list of students and detail view that displays student details for any selected student in the list.
- Develop example Django app that performs CSV and PDF generation forany models created in previous laboratory component.
- Develop a registration page for student enrolment as done in Module 2 but without page refresh using AJAX.

6.2 Ethical Hacking

Course Content

Web Application (In)security:
The Evolution of Web Applications, Common Web Application Functions, Benefits of Web Applications, Web Application Security, “This Site Is Secure”, The Core Security Problem: Users Can Submit; Arbitrary Input, Key Problem Factors, The New Security Perimeter, The Future of Web Application Security.

Core Defense Mechanisms: Handling User Access, Authentication, Session Management, Access Control, Handling User Input, Varieties of Input, Approaches to Input Handling, Boundary Validation, Multistep Validation and Canonicalization, Handling Attackers, Handling Errors, Maintaining Audit Logs, Alerting Administrators, Reacting to Attacks.

Attacking Authentication:
Authentication Technologies, Design Flaws in Authentication Mechanisms, Bad Passwords, Brute-Forcible Login, Verbose Failure Messages, Vulnerable Transmission of Credentials, Password Change Functionality, Forgotten Password Functionality, “Remember Me” Functionality, User Impersonation Functionality, Incomplete Validation of Credentials, Nonunique Usernames, Predictable Usernames, Predictable Initial Passwords, Insecure Distribution of Credentials, Implementation Flaws in Authentication, Fail-Open Login Mechanisms, Defects in Multistage Login Mechanisms, Insecure Storage of Credentials, Securing Authentication, Use Strong Credentials, Handle Credentials Secretively, Validate Credentials Properly, Prevent Information Leakage, Prevent Brute-Force Attacks, Prevent Misuse of the Password Change

Function, Prevent Misuse of the Account Recovery Function, Log, Monitor, and Notify.

Attacking Session Management:
The Need for State, Alternatives to Sessions, Weaknesses in Token Generation, Meaningful Tokens, Predictable Tokens, Encrypted Tokens, Weaknesses in Session Token Handling, Disclosure of Tokens on the Network, Disclosure of Tokens in Logs, Vulnerable Mapping of Tokens to Sessions, Vulnerable Session Termination, Client Exposure to Token Hijacking,

Liberal Cookie Scope, Securing Session Management, Generate Strong Tokens, Protect Tokens Throughout Their Life Cycle, Log, Monitor, and Alert.

Attacking Access Controls:

Common Vulnerabilities, Completely Unprotected Functionality, Identifier Based Functions, Multistage Functions, Static Files, Platform Misconfiguration, Insecure Access Control Methods, Attacking Access Controls, Testing with Different User Accounts, Testing Multistage Processes, Testing with Limited Access, Testing Direct Access to Methods, Testing Controls Over Static Resources, Testing Restrictions on HTTP Methods, Securing Access Controls, A Multilayered Privilege Model.

6.3 Data Science

Course Content

Introduction:

What is Data Science? Visualizing Data, matplotlib, Bar Charts, Line Charts, Scatterplots, Linear Algebra, Vectors, Matrices, Statistics, Describing a Single Set of Data, Correlation, Simpson's Paradox, Some Other Correlational Caveats, Correlation and Causation, Probability, Dependence and Independence, Conditional Probability, Bayes's Theorem, Random Variables, Continuous Distributions, The Normal Distribution, The Central Limit Theorem.

Hypothesis and Inference:

Statistical Hypothesis Testing, Example: Flipping a Coin, p-Values, Confidence Intervals, p-Hacking, Example: Running an A/B Test, Bayesian Inference, Gradient Descent, The Idea Behind Gradient Descent Estimating the Gradient, Using the Gradient, Choosing the Right Step Size, Using Gradient Descent to Fit Models, Minibatch and Stochastic Gradient Descent, Getting Data, stdin and stdout, Reading Files, Scraping the Web, Using APIs,. Example Using the Twitter APIs, Working with Data, Exploring Your Data, Using NamedTuples, Dataclasses, Cleaning and Munging, Manipulating Data, Rescaling, An Aside: tqdm, Dimensionality Reduction.

Machine Learnings

Modeling, What Is Machine Learning?, Overfitting and Underfitting, Correctness, The Bias-Variance Tradeoff, Feature Extraction and Selection, k-Nearest Neighbors, The Model, Example: The Iris Dataset, The Curse of Dimensionality, Naive Bayes, A Really Dumb Spam Filter, A More Sophisticated Spam Filter, Implementation, Testing Our Model,

Using Our Model, Simple Linear Regression, The Model, Using Gradient Descent, Maximum Likelihood Estimation, Multiple Regression, The Model, Further Assumptions of the Least Squares Model, Fitting the Model, Interpreting the Model, Goodness of Fit

Digression

Digression: The Bootstrap, Standard Errors of Regression Coefficients, Regularization, Logistic Regression, The Problem, The Logistic Function, Applying the Model, Goodness of Fit, Support Vector Machines

6.4 Internet of Things

Course Content

Emergence of IoT:

Introduction, Evolution of IoT, Enabling IoT and the Complex Interdependence of Technologies, IoT Networking Components, Addressing Strategies in IoT.

IoT Sensing and Actuation:

Introduction, Sensors, Sensor Characteristics, Sensorial Deviations, Sensing Types, Sensing Considerations, Actuators, Actuator Types, Actuator Characteristics.

IoT Processing Topologies and Types:

Data Format, Importance of Processing in IoT, Processing Topologies, IoT Device Design and Selection Considerations, Processing Offloading. IoT Connectivity Technologies: Introduction, IEEE 802.15.4, Zigbee, Thread, ISA100.11A, WirelessHART, RFID, NFC, DASH7, Z-Wave, Weightless, Sigfox, LoRa, NB- IoT, Wi- Fi, Bluetooth

IoT Communication Technologies:

Introduction, Infrastructure Protocols, Discovery Protocols, Data Protocols, Identification Protocols, Device Management, Semantic Protocols IoT Interoperability: Introduction, Taxonomy of interoperability, Standards, Frameworks IoT Interoperability: Introduction, Taxonomy of interoperability, Standards, Frameworks.